

ACID & BASE

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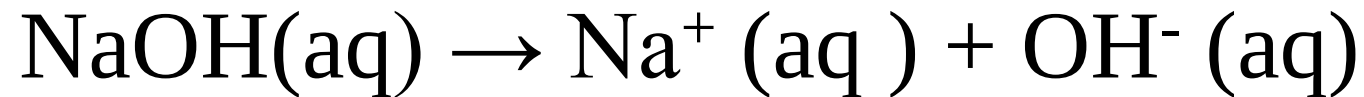
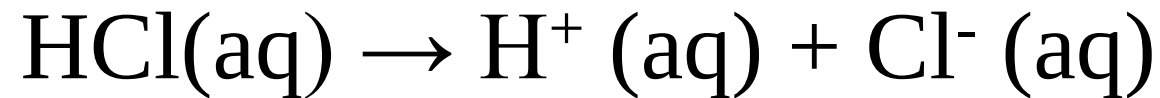
ARRHENIUS CONCEPT (1884)

Acid : An acid is any hydrogen containing compound that releases H^+ ions in water

Base : A base is a compound that releases OH^- ions in water.

Example:

HCl is an Arrhenius acid and **NaOH** is an Arrhenius base.



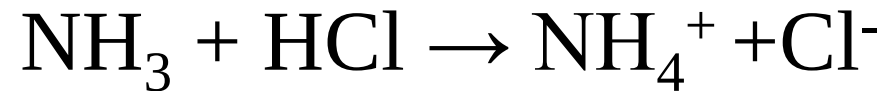
BONSTED–LOWRY CONCEPT (1923)

Acid: An acid is any molecule or ion that can donate a proton (H^+)

Base: A base is any molecule or ion that can accept a proton (H^+)

Example:

HCl and Ammonia, NH_3 : HCl gas reacts with ammonia (NH_3) to form solid NH_4Cl .



HCl is a proton donor and hence a **Bronsted acid**, while NH_3 is a proton acceptor and a **Bronsted base**.

Conjugate Acid Base Pairs

Conjugate Base : A conjugate base is the charged particle of the **original acid** after it donates its hydrogen ion (H^+)

Conjugate Acid : A conjugate acid is the charged particle formed when the **original base** gains a hydrogen ion (H^+)

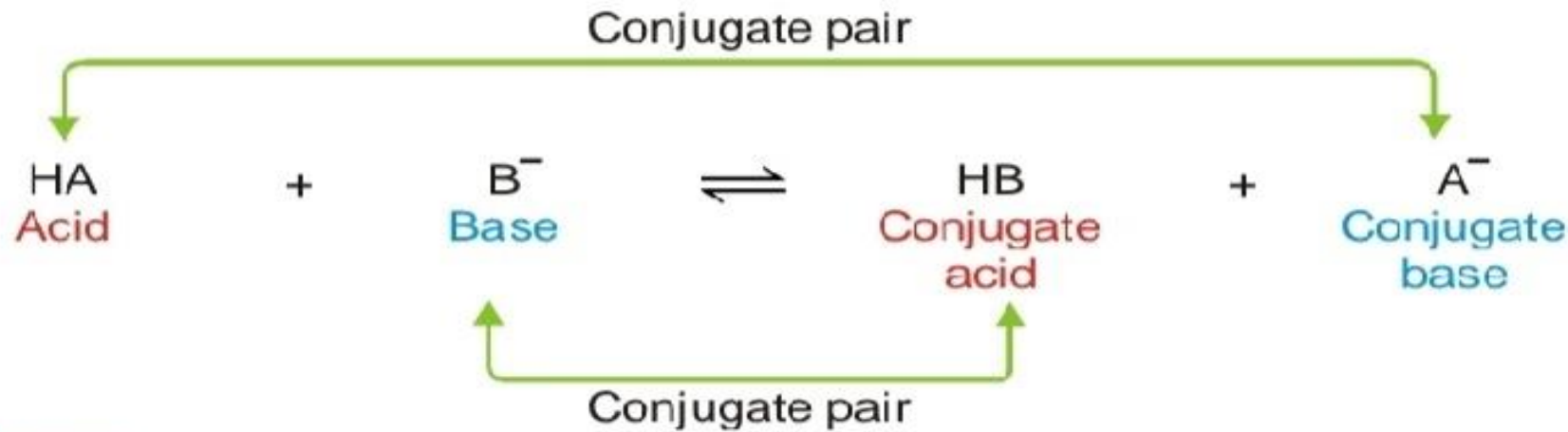
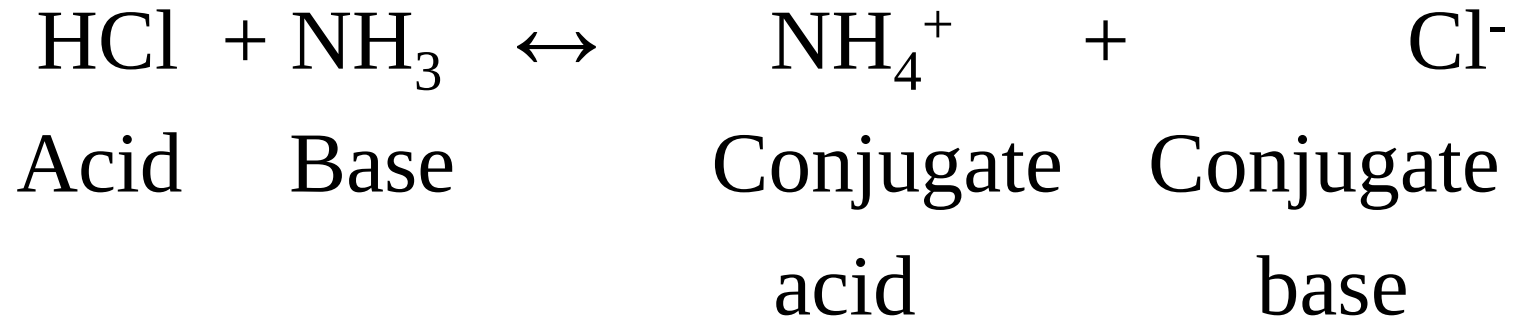


Figure:
Illustration of conjugate pairs.

Examples of Conjugate Acid-base pairs



In this case HCl (acid) and Cl^- (conjugate base) constitute one acid-base conjugate pair. Base NH_3 and the conjugate acid (NH_4^+) comprise the second acid-base conjugate pair.

Amphiprotic substances

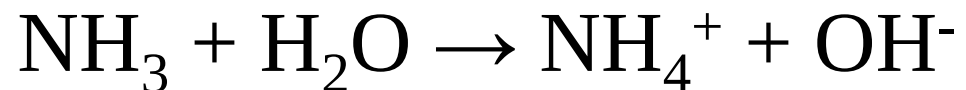
Molecules or ions that can behave both as Bronsted acid and base are called amphiprotic substances.

Example : With HCl, water acts as a base in accepting a proton from the acid.



Acid Base

However, water is an acid while donating a proton to ammonia.



Base Acid

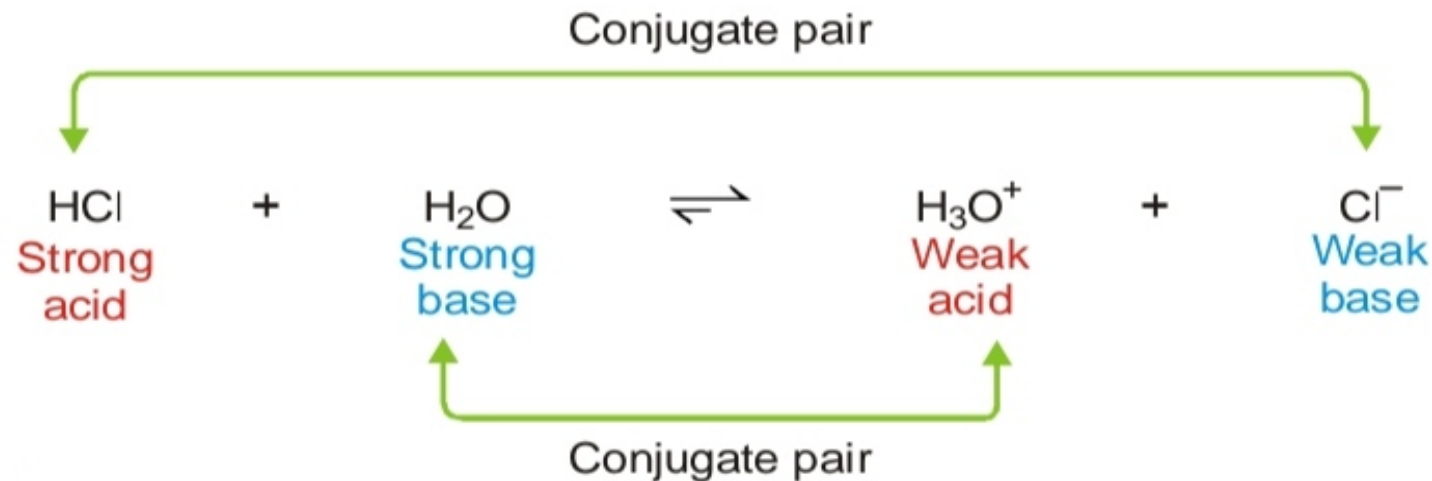
Strength Bronsted acids and bases

The strength of a Bronsted acid depends upon its tendency to donate a proton.
The strength of a Bronsted base depends on its ability to accept a proton.

A strong acid has a weak conjugate base

A strong base has a weak conjugate acid

Example :HCl is nearly 100% ionised in water. Its reaction with water can be depicted by the equation



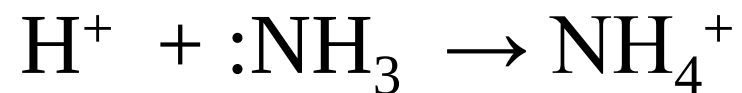
LEWIS CONCEPT(1930)

Acid : An acid is any species (molecule, radical or ions) which can accept an electron-pair.

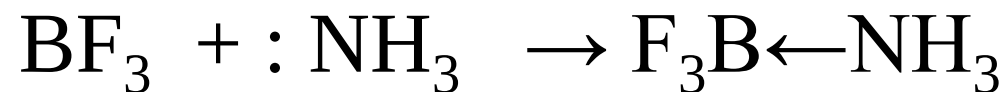
Base : A base is any species that can donate an electron-pair.

Example:

(1) Between H^+ and NH_3 : Proton (H^+) is a Lewis acid as it can accept an electron-pair. Ammonia molecule ($:\text{NH}_3$) has an electron-pair which it can donate and is a Lewis base.



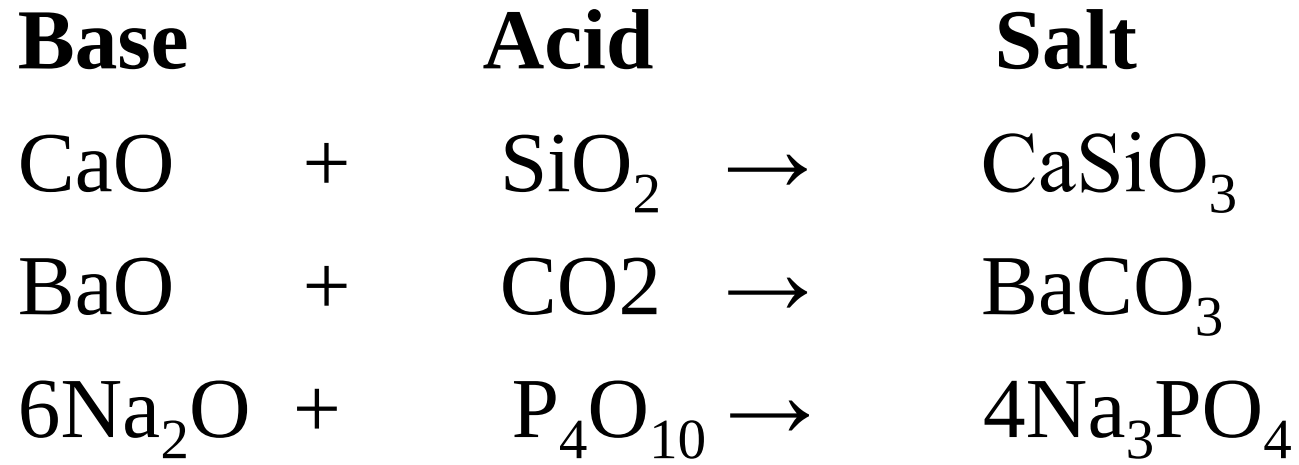
(2) Between BF_3 and NH_3 :



LUX-FLOOD CONCEPT

Acid: An acid is an oxide ion-acceptor.

Base: A base is an oxide ion-donor.



HSAB PRINCIPLE

Hard Acids

- Small ionic radii,
- High positive charge,
- Strongly solvated,
- Empty orbitals in the valence shell

Example: H^+ , Li^+ , Na^+ , K^+ , Be^{2+} , Mg^{2+} , Ca^{2+} , Sr^{2+} , Sn^{2+}

Al^{3+} , Ga^{3+} , In^{3+} , Cr^{3+} , Co^{3+} , Fe^{3+} ,

La^{3+} , Si^{4+} , Ti^{4+} , Zr^{4+} , Th^{4+} .

Soft Acids

- Large ionic radii,
- Low positive charge,
- Completely filled atomic orbitals

Example: Cu^+ , Ag^+ , Au^+ , Hg^+ , Cs^+ ,
 Hg^{2+} , Pd^{2+} , Cd^{2+} , Pt^{2+}

Hard Bases

- Small ionic radii,
- Highly electronegative,
- Weakly polarizable,
- Strongly solvated,

Example: H_2O , OH^- , F^- , Cl^- , CH_3COO^- , PO_4^{3-} , SO_4^{2-} , CO_3^{2-} ,
 NO_3^- , ClO_4^- , ROH , RO^- , R_2O , NH_3 ,

Soft Bases

- Large ionic radii,
- Intermediate electronegativity,
- highly polarizable

Example: RSH , RS^- , R_2S , I^- , CN^- , SCN^- , S_2O_3^- , R_3P , R_3As ,
 $(\text{RO})_3\text{P}$, RNC , CO , C_2H_4 , C_6H_6 , R^- , H^-

The pH OF SOLUTIONS

The negative of the base-10 logarithm (log) of the H^+ concentration. Mathematically it may be expressed as

$$\text{pH} = -\log [\text{H}^+]$$

where $[\text{H}^+]$ is the concentration of hydrogen ions in moles per litre.

pH Scale

The scale on which pH values are computed is called the pH scale.

Water dissociates to H^+ and OH^- ions to a very small degree so that we have the equilibrium.



We can write the equilibrium expression as

$$K = [\text{H}^+] [\text{OH}^-] / [\text{H}_2\text{O}]$$

Since water is so little dissociated, the concentration of undissociated molecules, $[\text{H}_2\text{O}]$, is presumed to be constant. Therefore we can write

$$[\text{H}^+] [\text{OH}^-] = K [\text{H}_2\text{O}] = \mathbf{K_w} = \text{a constant.}$$

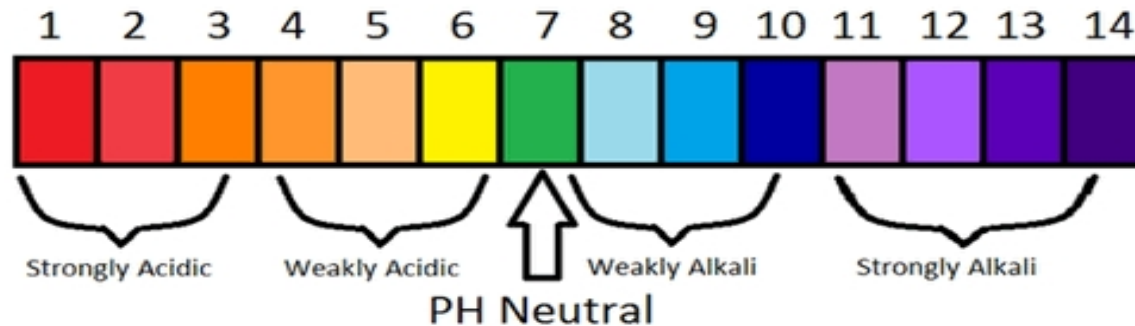
where $\mathbf{K_w}$ is called the water dissociation constant or the water ionisation constant.

Neutral solution $[H^+] = [OH^-]$

Acidic solution $[H^+] > [OH^-]$

Basic solution $[H^+] < [OH^-]$

All solutions having pH less than 7 are acidic and those with pH greater than 7 are basic.



As shown by the pH scale, pH decreases with the increase of $[H^+]$. The lower the pH, higher is the $[H^+]$ or acidity.

Relation between pH and pOH

$$\text{pOH} = -\log [\text{OH}^-]$$

$$\text{p}K_w = -\log K_w$$

Let us consider the log form of the expression

$$K_w = [\text{H}^+] [\text{OH}^-]$$

That is, $\log K_w = \log [\text{H}^+] + \log [\text{OH}^-]$

or $-\log K_w = -\log [\text{H}^+] - \log [\text{OH}^-]$

Thus $\text{p}K_w = \text{pH} + \text{pOH}$

Since $K_w = 1.0 \times 10^{-14}$

$$\text{p}K_w = -\log (1.0 \times 10^{-14}) = 14.00$$

Hence, for any aqueous solution at 25°C, pH and pOH add up to 14.00. That is,

$$\mathbf{pH + pOH = 14.00}$$

❖ **SOLVED PROBLEM 1. Calculate the pH of a 0.020 M Ba(OH)₂ solution.**

Solution: Barium hydroxide dissociates according to the equation



One molecule of barium hydroxide dissociates to give two OH⁻ ions. Therefore we have

$$[\text{OH}^-] = 0.040 \text{ M}$$

The pOH of the solution is, therefore,

$$\text{pOH} = -\log 0.040 = 1.40$$

$$\text{We have that } \text{pH} + \text{pOH} = 14.00$$

$$\therefore \text{pH} = 14.00 - 1.40 = 12.60$$

The pH of 0.020M Ba(OH)₂ solution is 12.60.

BUFFER SOLUTION

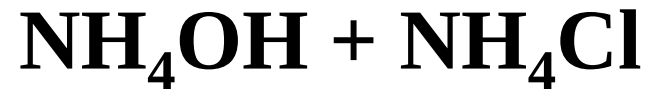
A buffer solution is a solution which resists changes in pH when a small amounts of acid or base is added.

Two common types of buffers are

(1) Acid Buffers : a weak acid together with a salt of the same acid with a strong base

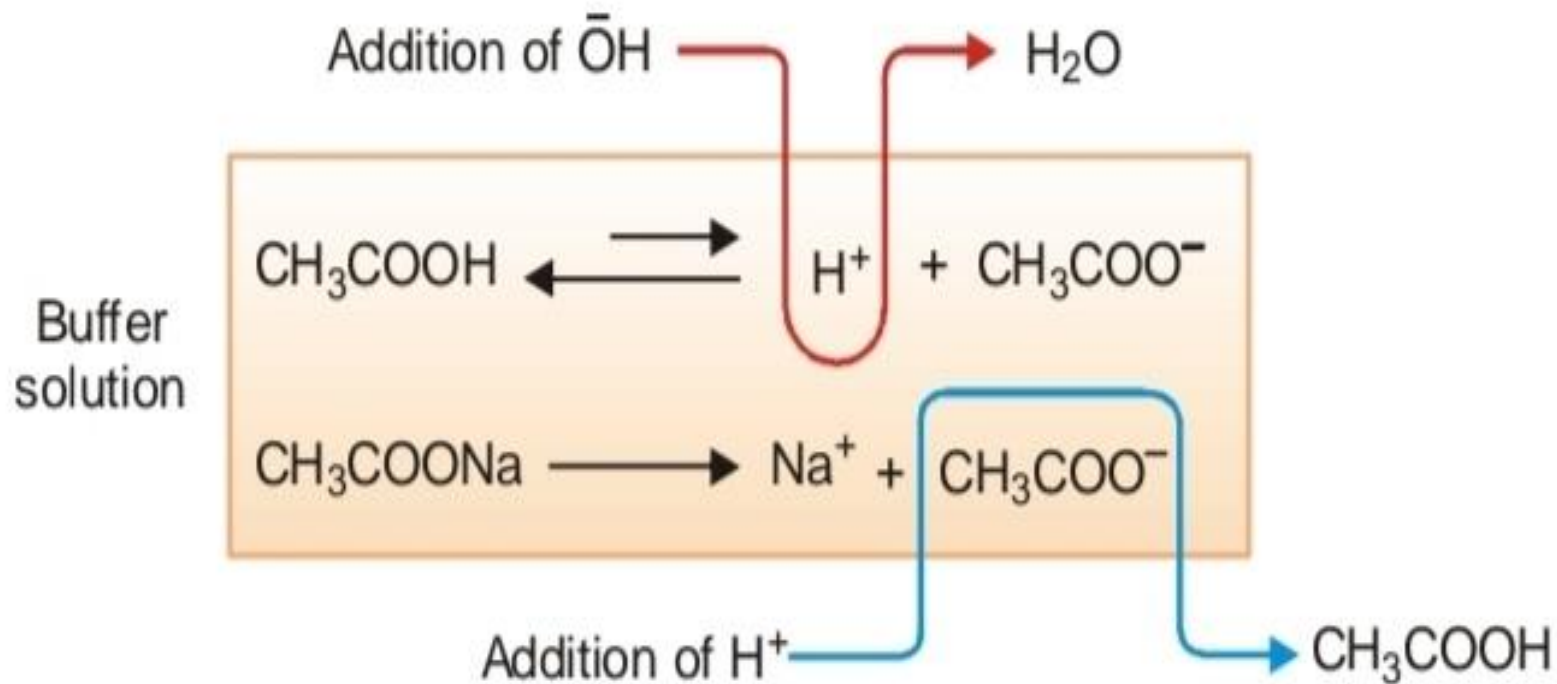


(2) Basic Buffers : a weak base and its salt with a strong acid

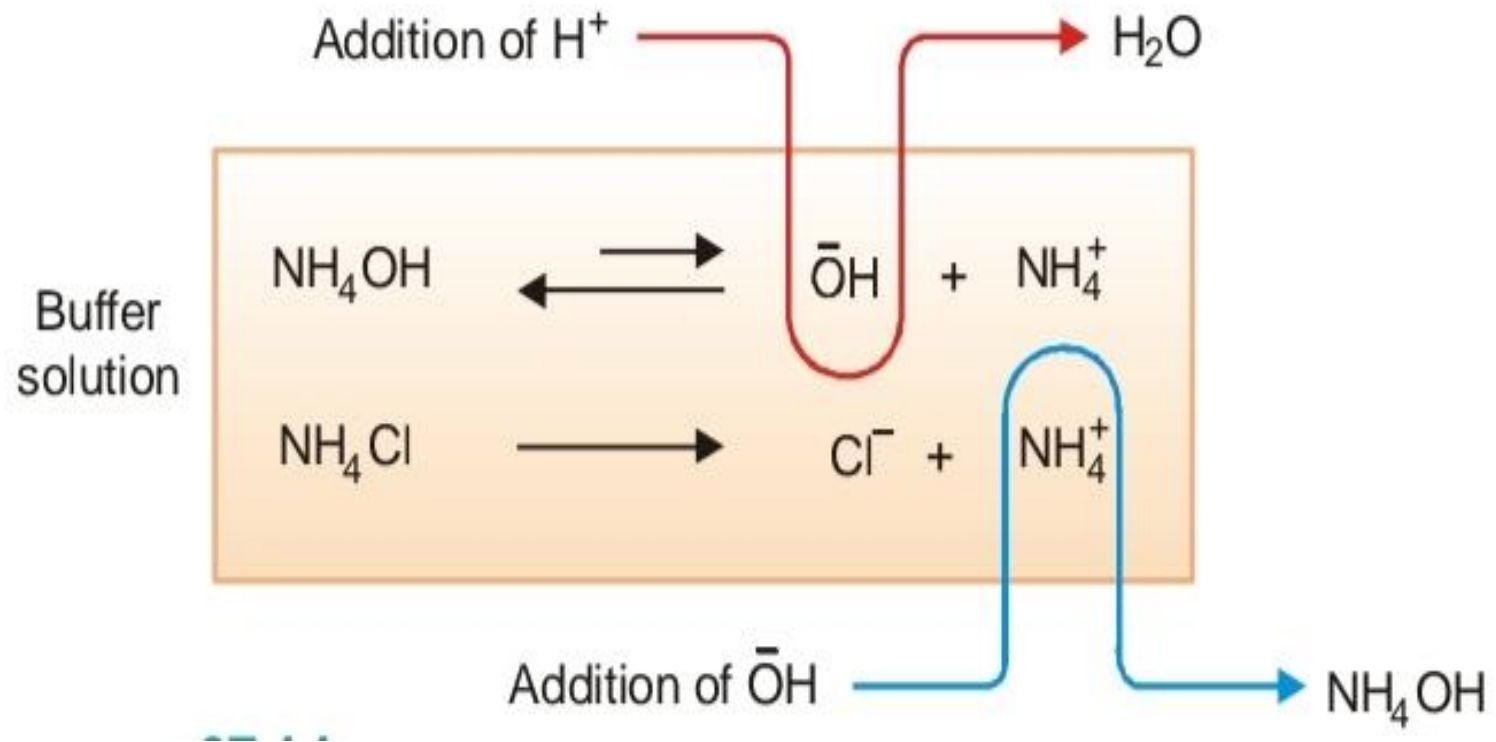


MECHANISM OF BUFFER SOLUTION

(1) Mechanism of Buffer action of an acid buffer



(2) Mechanism of Buffer action of a basic buffer.



ACID–BASE INDICATORS

An indicator is a substance which is used to determine the end point in a titration. In acid–base titrations, organic substances are generally used as indicators. They change their colour within a certain pH range.

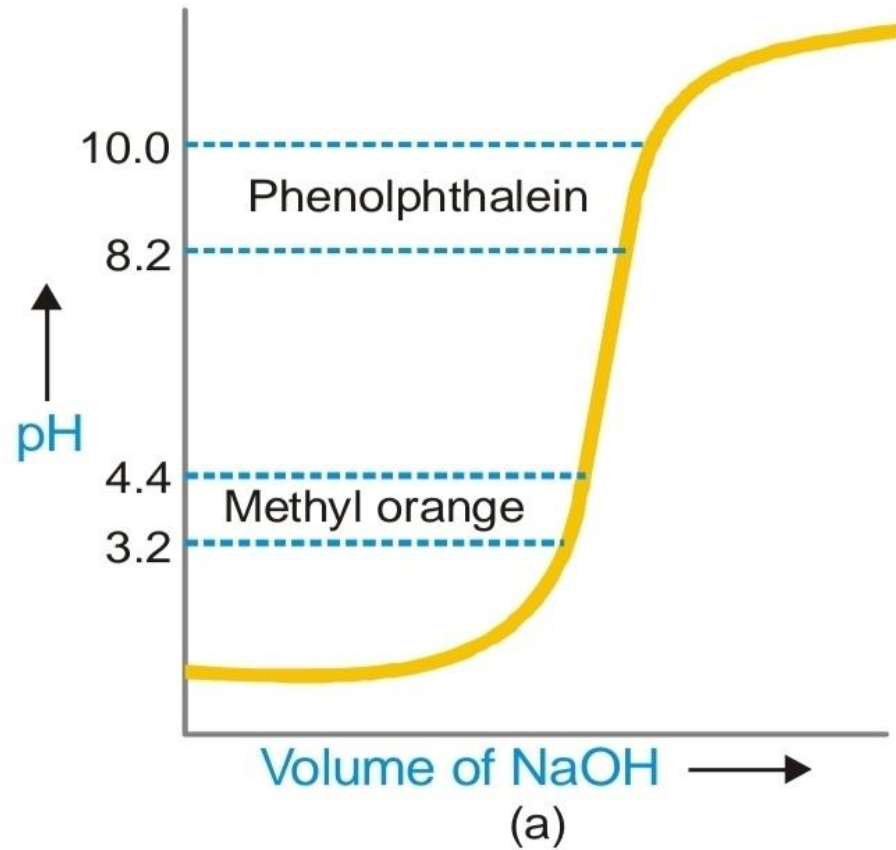
pH range of indicators

Indicator	Colour change (acid-base)		pH range
Methyl orange	red-orange	 → 	3.1 – 4.4
Methyl red	red-yellow	 → 	4.4 – 6.0
Litmus	red-blue	 → 	5.0 – 8.0
Bromothymol blue	yellow-blue	 → 	6.0 – 7.6
Phenolphthalein	colourless-pink	 → 	8.3 – 10.0

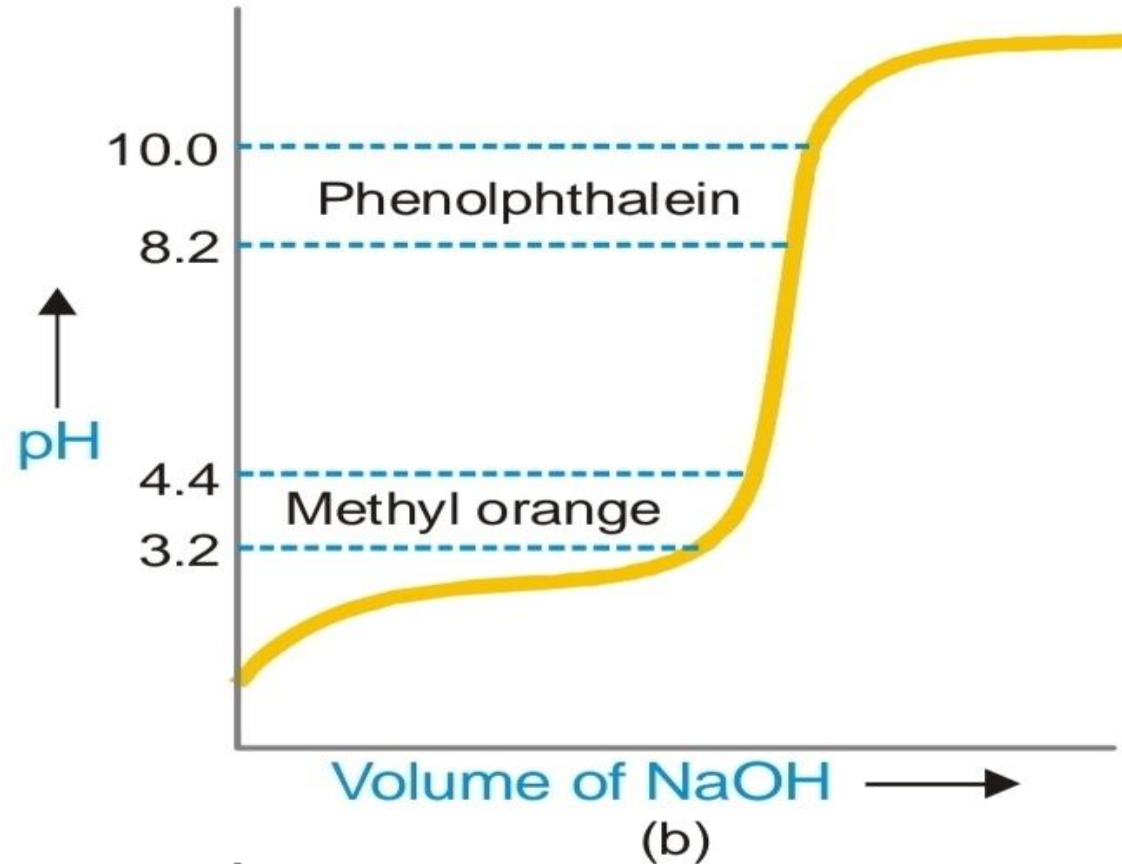
CHOICE OF A SUITABLE INDICATOR

(a) a strong acid with a strong base ($\text{NaOH} + \text{HCl}$):

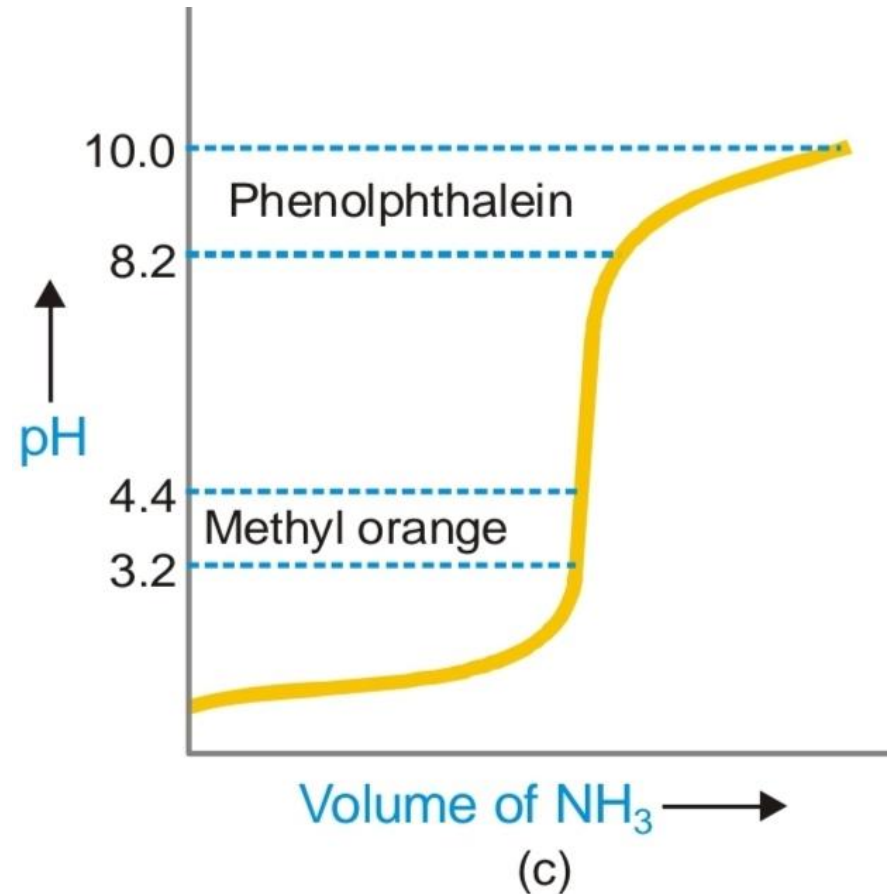
Both methyl orange and phenolphthalein are suitable indicators



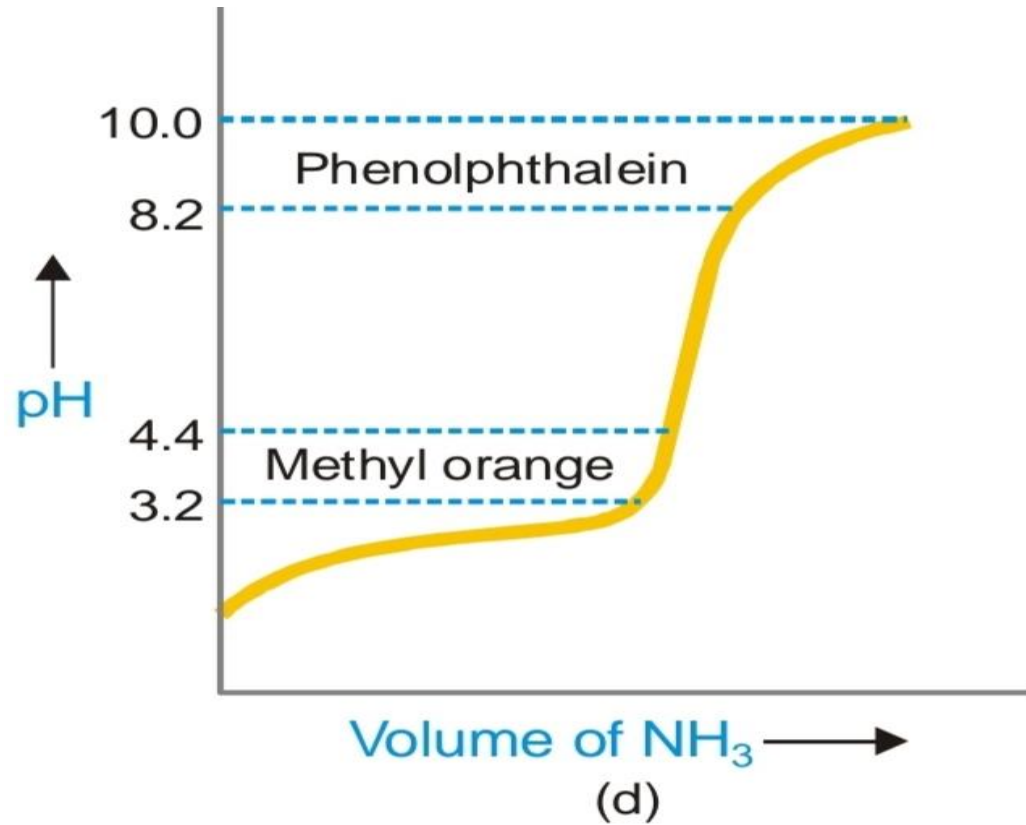
(b) a weak acid with a strong base($\text{NaOH} + \text{CH}_3\text{COOH}$):
Phenolphthalein is a suitable indicator



**(c) a strong acid with a weak base($\text{HCl} + \text{NH}_4\text{OH}$):
methyl orange and methyl red are suitable indicators.**



**(d) a weak acid with weak base($\text{CH}_3\text{COOH} + \text{NH}_4\text{OH}$):
no indicator is suitable**



Henderson Equation

The Henderson Hasselbalch equation is an approximate equation that shows the relationship between the pH or pOH of a solution and the pKa or pKb and the ratio of the concentrations of the dissociated chemical species. In order to use the equation, the acid dissociation constant must be known.

The pH of an acid buffer can be calculated from the dissociation constant, K_a , of the weak acid and the concentrations of the acid and the salt used.

The dissociation expression of the weak acid, HA, may be represented as



and

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

or

$$[\text{H}^+] = K_a \times \frac{[\text{HA}]}{[\text{A}^-]} \quad \dots(1)$$

$$[H^+] = K_a \times \frac{[\text{acid}]}{[\text{salt}]} \quad \dots(2)$$

where [acid] is the initial concentration of the added acid and [salt] that of the salt used.

Taking negative logs of both sides of the equation (2), we have

$$-\log[H^+] = -\log K_a - \log \frac{[\text{acid}]}{[\text{salt}]} \quad \dots(3)$$

But $-\log[H^+] = \text{pH}$ and $-\log K_a = \text{p}K_a$

Thus from (3) we have

$$\text{pH} = \text{p}K_a - \log \frac{[\text{acid}]}{[\text{salt}]} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

Hence $\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$

This relationship is called the **Henderson-Hasselbalch equation** or simply **Henderson equation**.

In a similar way, the Henderson-Hasselbalch equation for a basic buffer can be derived. This can be stated as :

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

SOLVED PROBLEM 1. Find the pH of a buffer solution containing 0.20 mole per litre CH_3COONa and 0.15 mole per litre CH_3COOH . K_a for acetic acid is 1.8×10^{-5} .

SOLUTION:

$$K_a = 1.8 \times 10^{-5}$$

$$\text{p}K_a = -\log(1.8 \times 10^{-5}) = 4.7447$$

$$\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

$$= 4.7447 + \log \frac{0.20}{0.15}$$

$$= 4.7447 + \log \frac{4}{3}$$

$$= 4.7447 + 0.6021 - 0.4771$$

$$= \mathbf{4.8697}$$

SOLVED PROBLEM 2. Calculate the pH of a buffer solution that is 0.250 M in formic acid, HCOOH, and 0.100 M in sodium formate, HCOONa. K_a for formic acid is 1.8×10^{-4} .

SOLUTION:

$$\begin{aligned}K_a &= 1.8 \times 10^{-4} \\pK_a &= -\log(1.8 \times 10^{-4}) \\&= 4 - \log 1.8 = 3.7447 \\pH &= pK_a + \log \frac{[\text{salt}]}{[\text{acid}]} \\&= 3.7447 + \log \frac{[0.100]}{[0.250]} \\&= 3.7447 - 0.3979 = \mathbf{3.3468}\end{aligned}$$

SOLVED PROBLEM 3. A buffer solution contains 0.015 mole of ammonium hydroxide and 0.025mole of ammonium chloride. Calculate the pH value of the solution. Dissociation constant of NH_4OH at the room temperature is 1.80×10^{-5} .

Solution:

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$\begin{aligned}\text{p}K_b &= -\log K_b \\ &= -\log (1.80 \times 10^{-5}) \\ &= 4.7447\end{aligned}$$

$$\begin{aligned}\text{pOH} &= 4.7447 + \log \frac{0.025}{0.015} \\ &= 4.7447 + 0.2218 \\ &= 4.9665\end{aligned}$$

$$\text{pH} + \text{pOH} = 14$$

$$\begin{aligned}\text{pH} &= 14 - 4.9665 \\ &= \mathbf{9.0335}\end{aligned}$$

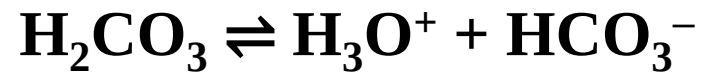
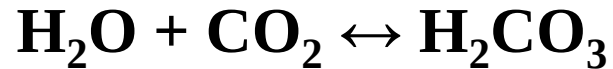
Importance of pH

- **Biological Uses** : In biological systems (saliva, stomach, blood) it is essential that the pH stays 'constant' in order for any processes to work properly

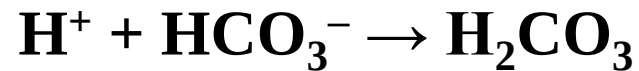
e.g. if the pH blood varies by 0.5 it can lead to unconsciousness and coma

Most enzymes work best at particular values

pH of human blood

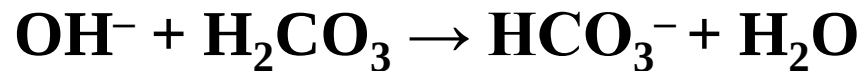


When an acid enters the bloodstream



This is to prevent the blood from becoming acidic

When a base enters the bloodstream



This is to prevent the blood from becoming basic

- **Other Uses** Many household and cosmetic product to control their pH values.

Shampoo : Buffer solutions counteract the alkalinity of the soap and prevent irritation

Baby lotion: Buffer solutions maintain a pH about 6 to prevent bacteria multiplying.

Others: washing Powder , eye drops, fizzy lemonade, hydroponic solutions.